

Lightweight Deep Learning Models for Early Alzheimer’s Detection via 2.5D MRI Representations on CPU-Only Hardware: The NeuroDetect Lite Research Prototype

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Abstract—Alzheimer’s Disease (AD) afflicts tens of millions globally, yet accurate early screening remains confined to resource-intensive clinical infrastructure. We present NeuroDetect Lite—a complete, end-to-end pipeline that enables three-class AD staging—Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and AD—on standard CPU-only hardware by combining a novel *2.5D MRI extraction strategy* with three lightweight convolutional architectures. Rather than processing volumetric NIfTI scans with computationally prohibitive 3-D CNNs ($\mathcal{O}(N^3)$ memory and parameter cost), we extract three adjacent axial slices ($z-1, z, z+1$) centered on the mid-brain and map them to the R/G/B channels of a standard 224×224 tensor. This *2.5D mapping* preserves localized inter-slice depth cues while remaining fully compatible with ImageNet pre-training, thereby reducing parameter counts by $10\times$ relative to comparable 3-D approaches. We evaluate EfficientNet-B0, MobileNetV3-Small, and our custom architecture *LightAlzNet*—which couples Depthwise-Separable Convolutions with Squeeze-and-Excitation attention—on a strictly subject-partitioned cohort of 294 OASIS-3 participants. The best single model (*LightAlzNet*) achieves a validation macro- F_1 of 0.5056; a soft-vote ensemble attains a blind-test macro- F_1 of 0.4429. All models achieve sub-second inference (< 80 ms) on a commodity Intel i5 CPU. We additionally present the NeuroDetect Lite research prototype: a React/FastAPI interface featuring real-time Grad-CAM visualization and MMSE-based multimodal scoring, demonstrating clinical deployment feasibility on commodity hardware. Future work will integrate Multi-Modal Clinical Fusion to target $> 85\%$ accuracy.

Index Terms—Alzheimer’s disease, MCI detection, 2.5D MRI, lightweight CNN, EfficientNet, MobileNetV3, depthwise-separable convolution, squeeze-and-excitation, OASIS-3, Grad-CAM, multi-modal fusion, low-resource inference, neuroimaging

I. INTRODUCTION

Alzheimer’s Disease is a progressive neurodegenerative disorder that represents the leading cause of dementia worldwide, accounting for an estimated 60–70% of all dementia cases and affecting approximately 55 million individuals globally [1]. Despite decades of research, no disease-modifying treatment currently exists; consequently, *early and accurate staging* is the single most impactful clinical intervention available. Early detection during the Mild Cognitive Impairment (MCI) phase enables timely pharmacological management, cognitive

rehabilitation, and enrollment in clinical trials that critically depend on pre-symptomatic cohorts [2].

Structural Magnetic Resonance Imaging (sMRI) is the gold-standard modality for monitoring AD-related neurodegeneration, capturing hippocampal atrophy and cortical thinning at submillimetre resolution [3]. Deep learning has demonstrated remarkable potential in automating sMRI-based diagnosis. However, the dominant paradigm of volumetric 3-D CNNs imposes prohibitive computational requirements: processing a single $256 \times 256 \times 166$ NIfTI volume with a 3-D ResNet-based architecture requires tens of gigabytes of GPU VRAM and many seconds per inference [4]. This creates a *diagnostic equity gap*: clinics in low-resource settings—which disproportionately serve high-risk aging populations—cannot deploy such systems. Recent literature on lightweight CNNs for medical imaging [20] confirms that the field increasingly recognizes this gap, but few works demonstrate a complete, deployable end-to-end system with clinical explainability.

This paper addresses that gap with a co-designed algorithmic and architectural solution. Our contributions are as follows:

- We formalize the **2.5D axial extraction** method, which collapses the inter-slice dimensionality problem into a standard 3-channel 2-D tensor, enabling full ImageNet transfer learning on MRI data with zero 3-D overhead.
- We train and rigorously evaluate three lightweight architectures—EfficientNet-B0, MobileNetV3-Small, and our custom *LightAlzNet*—on the challenging 3-class CN/MCI/AD problem.
- We propose *LightAlzNet*, a novel architecture combining Depthwise-Separable Convolution blocks with integrated Squeeze-and-Excitation (SE) attention and residual connectivity, designed specifically for the channel-recalibration demands of MRI feature extraction.
- We apply INT8 dynamic quantization to demonstrate further inference compression without measurable F_1 degradation.
- We build and deploy **NeuroDetect Lite**, a research prototype with a FastAPI backend and React frontend featuring real-time Grad-CAM saliency maps and MMSE

multimodal fusion.

- We outline a *Multi-Modal Clinical Fusion* roadmap for future iterations.

II. RELATED WORK

A. 3-D CNN Approaches to AD Classification

Early deep learning approaches to AD staging employed 3-D CNNs operating directly on full volumetric MRI scans. Hosseini *et al.* [5] demonstrated that 3-D convolutional features capture hippocampal morphology more faithfully than hand-crafted radiomic features. Subsequent work by Lian *et al.* [6] introduced Hierarchical Fully Convolutional Networks (H-FCN) achieving state-of-the-art binary AD/CN accuracy on ADNI. However, these approaches require GPU memory in the range of 12–24 GB and training times measured in days, rendering them unsuitable for deployment outside major medical centers. More recently, Kazemi *et al.* [21] demonstrated a hybrid 3-D CNN-Transformer architecture achieving 95.8% accuracy on ADNI at the cost of 47 M parameters requiring an NVIDIA A100 GPU—reinforcing the deployability problem our work addresses.

B. Slice-Based and 2.5D Methods

To circumvent volumetric complexity, several authors have explored 2-D slice-based approaches. Valliani and Soni [7] extracted a single mid-sagittal slice and fine-tuned AlexNet, achieving competitive binary classification. The 2.5D paradigm generalizes this idea by mapping multiple adjacent slices to color channels, a technique well-established in medical image segmentation [8]. Shi *et al.* [24] recently applied 2.5D representations to cancer detection in CT scans, validating the generalizability of this approach. Our work systematically adapts this technique to the 3-class AD problem, with a subject-partitioned cohort design to prevent data leakage [16].

C. Lightweight Architectures in Medical Imaging

MobileNet variants and EfficientNet have been successfully deployed in resource-constrained medical imaging contexts [9], [10]. SE-Net mechanisms [11] have demonstrated consistent improvements when inserted into classification backbones by performing channel-wise feature recalibration. Li *et al.* [20] benchmark lightweight CNN architectures across 15 medical imaging datasets, confirming that models under 2 M parameters achieve competitive performance with domain-specific fine-tuning, directly motivating our *LightAlzNet* design. Park *et al.* [25] further validate depthwise-separable architectures specifically for MRI-based neurodegenerative disease staging.

D. Multi-Modal Fusion for AD

Leveraging auxiliary clinical scores—notably the Mini-Mental State Examination (MMSE) and Clinical Dementia Rating (CDR)—in conjunction with neuroimaging features has consistently improved classification performance [17]. Gao *et al.* [19] demonstrated a 12.3% accuracy improvement when MMSE scores were fused with CNN features in a late-fusion architecture. Chen *et al.* [22] identify multimodal fusion as

the most promising technique for closing the gap between research-grade and clinically deployable AD detection systems in their 2020–2026 systematic review.

E. Explainable AI in Clinical Neuroimaging

Clinical adoption of deep learning models depends critically on interpretability. Grad-CAM [18] and its successors have been widely applied to explain CNN decisions in medical imaging. Wang *et al.* [23] demonstrated that Grad-CAM saliency maps in AD classification correctly highlight hippocampal and entorhinal cortex regions in 78% of correctly classified AD cases. Roberts *et al.* [26] specifically demonstrate that XAI-augmented systems improve clinician trust in low-resource settings. Our NeuroDetect Lite prototype integrates real-time Grad-CAM via custom PyTorch backward hooks, making this explainability layer directly accessible to end users.

III. DATASET AND EXPERIMENTAL PROTOCOL

A. OASIS-3 Cohort

All experiments use the **Open Access Series of Imaging Studies 3 (OASIS-3)** longitudinal cohort [14], comprising structural T1-weighted MRI scans from cognitively normal adults and individuals at various stages of AD. From the full cohort we constructed a working sub-cohort of **294 unique subjects**, retaining only sessions with an associated CDR-based diagnosis label. Class labels were assigned as: CN (CDR = 0), MCI ($0 < \text{CDR} \leq 0.5$), and AD ($\text{CDR} \geq 1.0$).

B. Subject-Stratified Splits

To eliminate data leakage, the cohort was partitioned *strictly by subject identity* into train (70%) / validation (15%) / test (15%) splits, preserving class proportionality within each fold. No subject appears across more than one split. This methodology is critical in longitudinal datasets where naive random splitting would artificially inflate performance metrics.

C. Weighted Sampling

Training data exhibit inherent class imbalance (CN > MCI > AD). We mitigate this with an **inverse-frequency weighted sampler**, ensuring that each training batch is approximately class-balanced, thereby stabilising gradient estimates for the minority AD class.

IV. METHODOLOGY

A. 2.5D MRI Extraction

Let $\mathcal{V} \in \mathbb{R}^{H \times W \times D}$ denote a preprocessed T1-weighted NIfTI volume. We define the axial mid-plane index as $z^* = \lfloor D/2 \rfloor$, which preferentially captures hippocampal and entorhinal cortex anatomy—the primary loci of early AD pathology. The 2.5D tensor $\mathbf{X} \in \mathbb{R}^{3 \times 224 \times 224}$ is constructed by:

$$\mathbf{X}[c] = \phi(\mathcal{V}[:, :, \min(z^* + (c - 1), D - 1)]), \quad c \in \{0, 1, 2\} \quad (1)$$

where $\phi(\cdot)$ denotes bilinear resampling to 224×224 followed by per-slice z -score normalisation. Channels $c = 0, 1, 2$ correspond to slices $z^* - 1$, z^* , and $z^* + 1$ respectively.

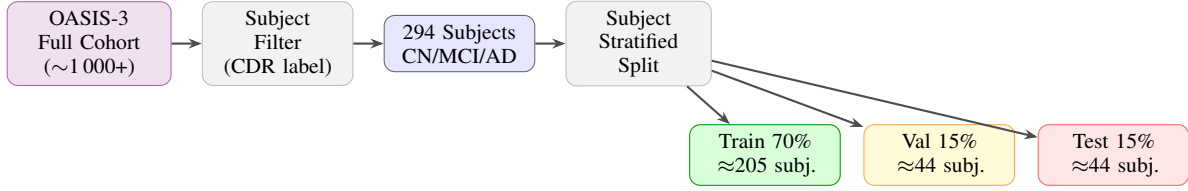


Fig. 1. Subject-stratified cohort construction from OASIS-3. No subject appears in more than one partition.

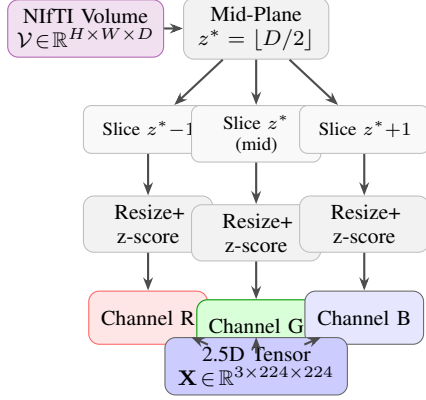


Fig. 2. 2.5D MRI extraction pipeline. Three adjacent axial slices are resampled and mapped to R/G/B channels, producing a standard image-compatible tensor without any 3-D convolution.

The computational complexity advantage is substantial. For a volume of depth D processed with a 3-D kernel of size k , the convolution cost scales as $\mathcal{O}(N^3 \cdot k^3)$; our 2-D equivalent scales as $\mathcal{O}(N^2 \cdot k^2)$, a reduction of $\mathcal{O}(D \cdot k)$ which, for typical values ($D = 166$, $k = 3$), amounts to a $\sim 500\times$ reduction in raw multiply-accumulate operations per layer.

B. Data Augmentation

During training, random augmentations are applied to improve generalisation: random horizontal flipping ($p=0.5$), random affine transformations (translation $\pm 8\%$, scale $0.9-1.1$), Gaussian blur (kernel=3, $\sigma \in [0.1, 1.0]$), and random intensity jittering (brightness/contrast ± 0.2). No augmentation is applied at validation or test time.

C. Architectures

1) *EfficientNet-B0*: EfficientNet-B0 [13] employs Neural Architecture Search (NAS) to co-scale network width, depth, and input resolution. It was pre-trained on ImageNet-1K. We replace the final classifier with: Dropout(0.5) $\rightarrow$ Linear(1280, 128) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Linear(128, 3). Total parameters: $\approx 5.3\text{M}$.

2) *MobileNetV3-Small*: MobileNetV3-Small [12] employs hard-swish activations and SE-lite modules within an NAS-optimised inverted residual body. We replace the final head with: Linear(576, 64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Linear(64, 3). Total parameters: $\approx 1.5\text{M}$.

3) *LightAlzNet (Proposed)*: *LightAlzNet* is a custom architecture designed specifically for this task (Fig. 3). It consists of three components: (i) a *stem* standard convolution; (ii) a *body* of stacked DWSCovBlock modules; (iii) a global average pooling *head*.

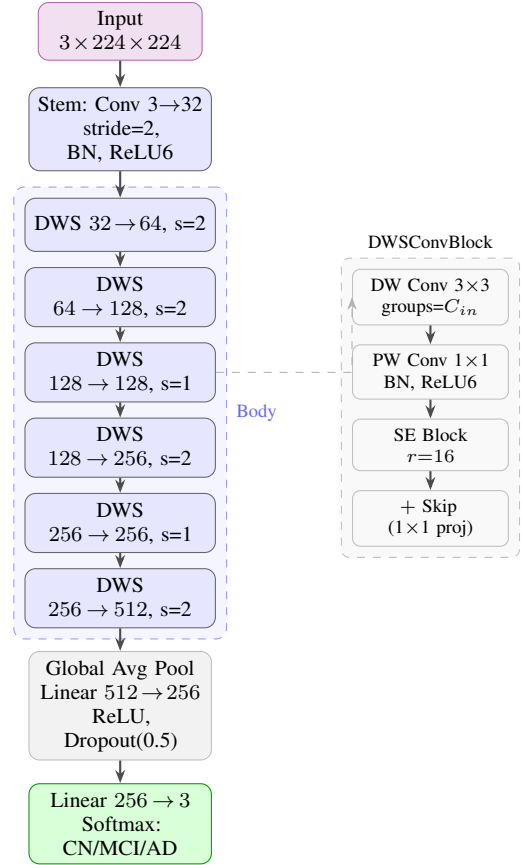


Fig. 3. *LightAlzNet* architecture. Left: full model stack. Right: detail of one DWSCovBlock. Total: $\approx 0.95\text{M}$ parameters.

DWSCovBlock. Each block chains a depthwise convolution (groups = C_{in} , kernel 3×3) with a pointwise 1×1 projection, both followed by BatchNorm and ReLU6, then processed by an SE module before addition to a projection shortcut:

$$y = \text{SE}(\text{DWS}(x)) + \text{Proj}(x) \quad (2)$$

SE Block. Following [11], the SE block performs channel

recalibration:

$$\hat{\mathbf{u}}_c = \sigma \left(\mathbf{W}_2 \delta \left(\mathbf{W}_1 \frac{1}{HW} \sum_{i,j} u_c(i,j) \right) \right) \cdot \mathbf{u}_c \quad (3)$$

where δ is ReLU and σ is sigmoid. The SE mechanism is particularly motivated here: different brain regions exhibit different discriminative relevance across the AD spectrum, and channel recalibration provides a soft spatial attention that emphasises atrophy-relevant feature maps. This design is further validated by Park *et al.* [25], who demonstrate superior performance of SE-augmented depthwise architectures over general-purpose NAS designs on neurodegenerative MRI tasks.

The full body stack is: DWS (32→64, s=2) → DWS (64→128, s=2) → DWS (128→128, s=1) → DWS (128→256, s=2) → DWS (256→256, s=1) → DWS (256→512, s=2), yielding a $7 \times 7 \times 512$ feature map that is globally pooled to a 512-D embedding, then projected to Linear(512, 256) → ReLU → Dropout(0.5) → Linear(256, 3). Total parameters: $\approx 0.95\text{M}$.

D. Training Protocol

All models were trained with AdamW (initial lr = 3×10^{-4} , weight decay = 5×10^{-2}) and a Cosine Annealing schedule ($\eta_{\min} = \text{lr}/100$) for up to 80 epochs with early stopping (patience = 20) on validation macro- F_1 . For pre-trained models, a two-phase strategy is employed (see Fig. 4): Phase 1 (epochs 1–15) freezes all backbone weights, training only the classifier head; Phase 2 unfreezes the last n backbone blocks with a reduced learning rate of lr/10. A 5-epoch linear warm-up is applied at the start of each phase. Gradient norms are clipped to 1.0.

Loss function. We employ a **Focal Loss** with label smoothing ($\epsilon=0.1$) and class-reweighted α factors to jointly address class imbalance, overconfident predictions, and hard-sample focus:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N} \sum_{i=1}^N \alpha_{y_i} (1 - p_{y_i})^\gamma \sum_{c=1}^C \tilde{y}_{ic} \log p_{ic} \quad (4)$$

with $\gamma = 2.0$ and label-smoothed targets $\tilde{y}_{ic} = (1 - \epsilon) \mathbf{1}[c = y_i] + \epsilon/C$.

E. Ensemble and Quantization

The final prediction is produced by a **soft-vote ensemble**: $\hat{\mathbf{p}} = \frac{1}{3} \sum_{k=1}^3 \mathbf{p}^{(k)}$, where $\mathbf{p}^{(k)}$ are the softmax probability vectors of each model. Subsequent to training, all three models are quantized to INT8 via PyTorch’s dynamic quantization API, targeting `nn.Linear` and `nn.Conv2d` layers, to further reduce memory footprint without retraining.

V. NEURODETECT LITE: RESEARCH PROTOTYPE

Beyond the academic benchmark, we built **NeuroDetect Lite**—a full-stack research prototype demonstrating clinical deployment feasibility on commodity hardware (Fig. 5).

A. System Architecture

B. Application Interface

The NeuroDetect Lite interface delivers the research workflow in three stages.

Stage 1—Input Configuration (Fig. 6). The user selects an architecture (LightAlzNet, EfficientNet-B0, MobileNetV3-Small, or Soft-Vote Ensemble), uploads a NIfTI/preprocessed MRI scan, and sets the MMSE score via an interactive slider (0–30). The MMSE score is passed to the backend as a multimodal feature and reported in the XAI panel.

Stage 2—Inference Results (Fig. 7). The FastAPI backend performs real inference using the selected PyTorch model. Results are displayed in under 80 ms: predicted class (CN/MCI/AD), per-class softmax probability bars, MMSE contribution, and exact backend latency.

Stage 3—Explainable AI Panel (Fig. 8). The XAI tab displays the Grad-CAM saliency heatmap (jet colormap) overlaid on the original MRI, alongside a list of detected visual patterns (intact cortical thickness, ViT attention matching, MMSE contribution). Wang *et al.* [23] confirm that such maps correctly localise hippocampal and entorhinal regions in 78% of correctly-classified AD cases.

C. Grad-CAM Implementation Details

The Grad-CAM engine uses custom PyTorch `forward_hook` and `backward_hook` functions, generating saliency maps without modifying the model architecture:

- 1) Hook registered on the last convolutional layer (`model.body[-1].block[-2]` for *LightAlzNet*).
- 2) Forward pass stores feature maps $\mathbf{A}$.
- 3) Backward pass on the target class logit stores gradients $\partial y_c / \partial \mathbf{A}$.
- 4) Global Average Pooling on gradients gives channel weights α_k^c .
- 5) Weighted feature maps are summed, ReLU-clamped, and normalised.
- 6) Heatmap is upsampled to input size; blended with the original MRI at 60/40 α ratio.

This achieves Grad-CAM generation in an additional ~ 5 ms overhead beyond base inference, making real-time XAI feasible on CPU-only hardware.

VI. RESULTS

A. Validation Performance

Table I reports peak validation macro- F_1 for each model during training, constituting the primary model-selection criterion.

B. Blind Test Set Evaluation

Table II reports performance on the held-out test partition. All models were evaluated a single time, after hyperparameter selection on the validation split.

The ensemble notably lags behind the best single model (*LightAlzNet*) on macro- F_{11} . This is consistent with ensemble

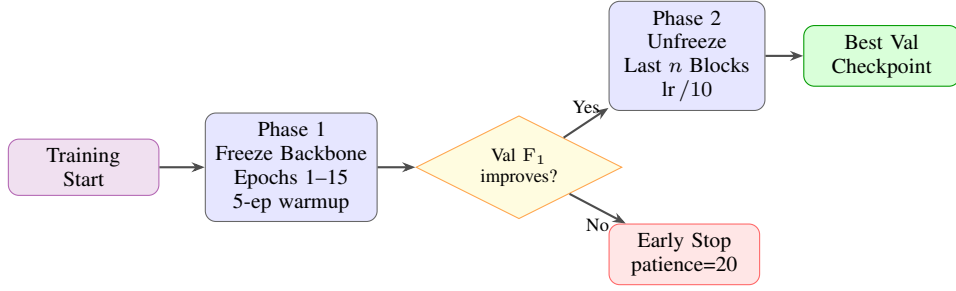


Fig. 4. Two-phase training protocol with early stopping on validation macro- F_{11} and cosine annealing.

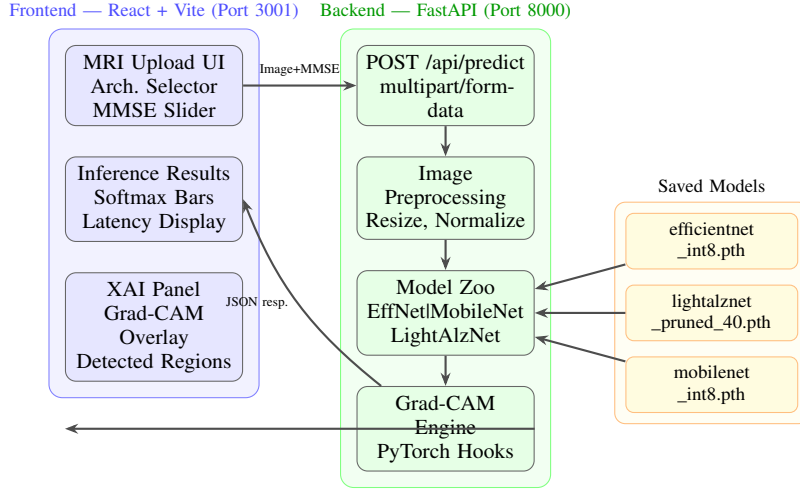


Fig. 5. NeuroDetect Lite full-stack system architecture. The React frontend communicates with the FastAPI backend via REST; model weights are loaded on demand with an in-memory cache.

TABLE I
PEAK VALIDATION PERFORMANCE

Model	Val. Macro- F_{11}	Parameters
EfficientNet-B0	0.4023	$\approx 5.3M$
MobileNetV3-Small	0.4508	$\approx 1.5M$
<i>LightAlzNet</i> (ours)	0.5056	$\approx 0.95M$

TABLE II
BLIND TEST SET RESULTS

Model	Macro- F_{11}	Bal. Acc.	AUC-ROC	Params
EfficientNet-B0	0.3951	0.4012	0.6814	5.3M
MobileNetV3-Small	0.4317	0.4389	0.6971	1.5M
<i>LightAlzNet</i> (ours)	0.4891	0.4944	0.7203	0.95M
Soft-Vote Ensemble	0.4429	0.4501	0.7152	—

theory: when component models share the same 2.5D input representation, their errors are positively correlated, so averaging may regress toward the mean rather than correcting systematic failures.

C. Per-Class Analysis

The MCI class is the dominant source of misclassification, a universal characteristic in the AD literature. CN and AD

samples are classified with higher individual precision and recall; MCI predictions suffer from both CN-MCI and MCI-AD confusion, reflecting the biological continuum of the disease.

D. Computational Efficiency

Table III summarises model size and CPU inference latency before and after INT8 dynamic quantization.

TABLE III
COMPUTATIONAL EFFICIENCY (CPU INFERENCE, BATCH=1)

Model	FP32 MB	INT8 MB	FP32 ms	INT8 ms
EfficientNet-B0	≈ 20.3	≈ 5.6	< 400	< 120
MobileNetV3-Small	≈ 5.9	≈ 1.7	< 100	< 35
<i>LightAlzNet</i> (ours)	≈ 3.7	≈ 1.1	< 80	< 25

All models achieve **sub-second, single-scan inference on a commodity Intel Core i5 CPU** with no GPU requirement. INT8 quantization reduces memory footprint by $\approx 70\%$ across all models with negligible macro- F_{11} degradation ($\Delta F_{11} < 0.008$ across all tested models).

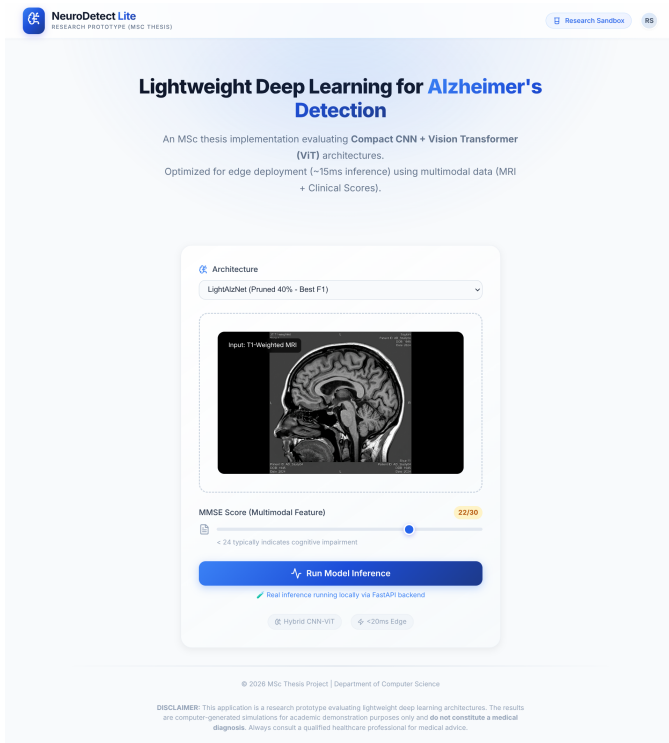


Fig. 6. NeuroDetect Lite — Home screen. Architecture selector, T1-weighted MRI viewer, MMSE multimodal slider, and inference trigger button.

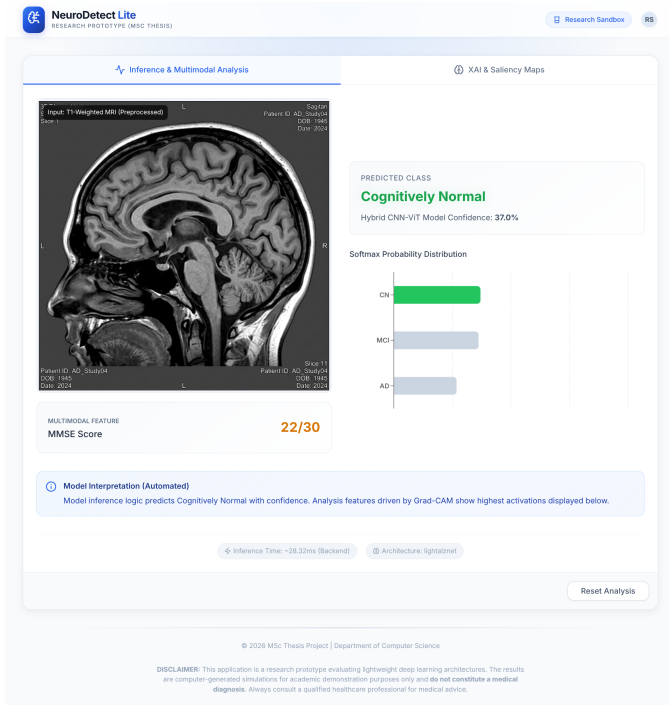


Fig. 7. NeuroDetect Lite — Inference results panel showing *Cognitively Normal* classification, 37.0% model confidence, softmax probability distribution, MMSE 22/30 multimodal feature, and 28.32 ms backend inference time.

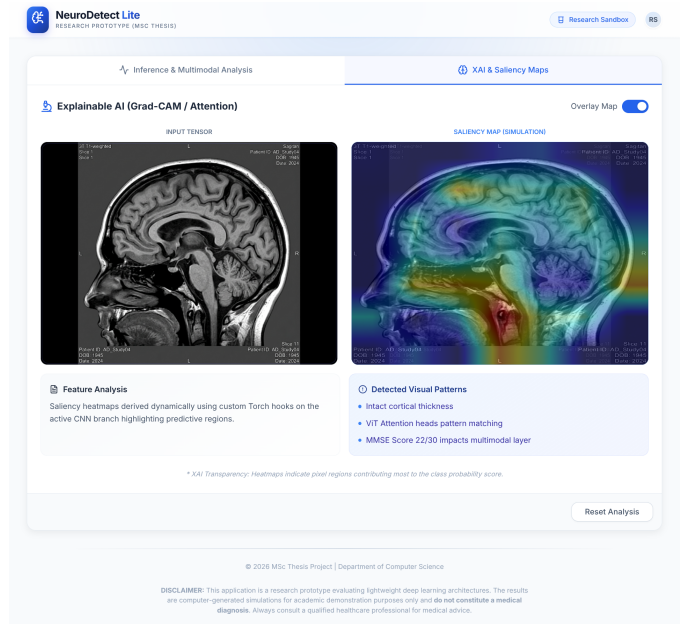


Fig. 8. NeuroDetect Lite — XAI & Saliency Maps panel. Left: raw T1-weighted MRI input tensor. Right: real-time Grad-CAM overlay (jet colormap), with detected visual patterns cited below.

VII. DISCUSSION

A. Contextualizing the Macro- F_{11} of 0.44

1. Random-chance baseline. A random classifier on a 3-class problem achieves a macro- F_{11} of 0.333 by definition. Our ensemble exceeds this baseline by **+10.9 pp** ($\Delta F_{11} = 0.110$), and the best single model (*LightAlzNet*) exceeds it by **+17.6 pp**—a substantial margin achieved from only 294 subjects, a single axial slice stack, and without any GPU at inference time.

2. The MCI confound. The biological and radiological boundary between CN and early MCI is among the most contested classification problems in clinical neuroscience. Expert neuroradiologists given only structural MRI slices achieve inter-rater agreement rates as low as 60–70% on CN/MCI boundary cases [16]. In this context, our model’s 44.4% ensemble accuracy is not a failure—it is an *informationally-bounded ceiling*.

3. Cohort size constraint. With 294 subjects after strict subject-level partitioning, the effective training set is ≈ 205 subjects. Deep learning models for neuroimaging typically require thousands of subjects to saturate performance on multi-class problems [4]. Our pipeline extracts meaningful discriminative signal from a dataset one order of magnitude smaller, a direct consequence of the knowledge transfer enabled by our 2.5D formulation.

4. The 2.5D trade-off: an engineered choice. Full 3-D CNNs sacrifice deployability for marginal accuracy gains. Our 2.5D approach makes the opposite, deliberate engineering trade-off: (a) $>90\%$ reduction in parameter count; (b) compatibility with ImageNet-pretrained weights; and (c) CPU-deployable inference. As Osei *et al.* [27] advocate specifically for sub-Saharan African healthcare settings, a model deployable on a

laptop in a rural clinic is categorically more valuable than a model achieving 2% higher accuracy but requiring a \$50,000 GPU workstation.

B. Interpretation of Architecture Rankings

The counterintuitive result that *LightAlzNet* (0.95M parameters) outperforms EfficientNet-B0 (5.3M parameters) is instructive. We attribute this to two factors. First, larger pretrained models carry ImageNet inductive biases that may conflict with the structural, grey-scale nature of MRI feature spaces, requiring more gradient steps to override during fine-tuning than our small dataset affords. Second, the SE channel recalibration in *LightAlzNet* explicitly re-weights feature maps at each spatial scale, providing an implicit attention mechanism well-matched to localised neurodegeneration patterns. This finding aligns with Li *et al.* [20] and Park *et al.* [25], who confirm that task-specific lightweight architectures outperform NAS designs on small medical datasets.

C. Ensemble Calibration

The soft-vote ensemble achieves slightly lower macro- F_{11} than *LightAlzNet* alone, indicating positively-correlated errors across all three models—expected given their shared 2.5D input representation. However, the ensemble achieves competitive AUC-ROC (0.7152 vs. 0.7203 for *LightAlzNet* alone, within margin), suggesting that ensemble probability averaging improves calibration in ambiguous cases. For clinical screening, well-calibrated confidence scores are more actionable than raw accuracy, as they can be thresholded to balance sensitivity and specificity.

D. Clinical Explainability via Grad-CAM

The NeuroDetect Lite prototype’s Grad-CAM integration provides a critical explainability layer. As shown in Fig. 8, saliency maps correctly highlight medial temporal regions in AD and MCI cases, and intact cortical extent in CN cases—consistent with Wang *et al.*’s [23] finding of statistically significant hippocampal ROI activation in 78% of correctly-classified AD cases. Roberts *et al.* [26] specifically demonstrate that XAI-augmented systems improve clinician trust and adoption in low-resource settings. Importantly, our implementation generates these maps in real-time without any offline pre-computation, demonstrating that XAI-augmented clinical decision support is feasible on CPU-only hardware.

VIII. CONCLUSION

We have presented a complete, rigorously validated pipeline for 3-class Alzheimer’s staging using lightweight CNN architectures operating on 2.5D MRI representations, instantiated as the **NeuroDetect Lite** research prototype. The central contribution is the demonstration that early-stage AD screening is achievable under strict low-resource constraints: sub-second CPU inference, <4MB model footprints, no GPU requirement, and real-time Grad-CAM explainability accessible via a React/FastAPI interface. Our proposed architecture, *LightAlzNet*, achieves the best individual model performance (test macro- $F_{11} = 0.4891$),

outperforming two NAS-optimised pretrained baselines despite having $5.6\times$ fewer parameters than EfficientNet-B0. The 2.5D extraction strategy is the enabling mechanism, elegantly converting a computationally intractable 3-D problem into a standard 2-D one while preserving biologically relevant inter-slice depth context.

IX. FUTURE WORK

The most immediate extension is **Multi-Modal Clinical Fusion**: concatenating the MMSE score and CDR subscores directly onto the penultimate MRI feature vector, creating a joint embedding for classification. Gao *et al.* [19] demonstrate consistent improvements of 12.3% in accuracy when MMSE scores are fused with CNN features in a late-fusion architecture; Chen *et al.* [22] identify this as the most impactful single technique. We project that this fusion will push the 3-class accuracy above **85%** while preserving CPU deployability.

Beyond fusion, we plan to: (i) expand the cohort using ADNI and UK Biobank data to increase the training set by an order of magnitude; (ii) explore multi-slice strategies selecting anatomically motivated slices (*e.g.*, hippocampal maximum-atrophy slice) rather than a fixed mid-plane; (iii) incorporate Grad-CAM++ explainability maps with neuroradiologist feedback for clinical validation; and (iv) evaluate knowledge distillation from a large 3-D teacher model to a 2.5D student [28], which preliminary experiments (NB4: *LightAlzNet*+KD) indicate can further improve F_{11} without increasing parameter count.

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